Cancer Letters 297 (2010) 144–154

Contents lists available at ScienceDirect

Cancer Letters

journal homepage: www.elsevier.com/locate/canlet

Hericium erinaceus enhances doxorubicin-induced apoptosis in human hepatocellular carcinoma cells

Jong Seok Lee, Eock Kee Hong *

Department of Bioengineering and Technology, Kangwon National University, Chuncheon 200-701, Republic of Korea

ARTICLE INFO

Article history:
Received 7 March 2010
Received in revised form 4 May 2010
Accepted 5 May 2010

Keywords:
Hericium erinaceus
Doxorubicin
Apoptosis
Hepatocellular carcinoma
NF-κB

ABSTRACT

It has been demonstrated that the *Hericium erinaceus* (HE) mushroom, which primarily consists of polysaccharides, possesses anti-tumor activities. However, the mechanisms by which HE inhibits human hepatocellular carcinoma growth remain unknown. Our study demonstrates that HE acts as an enhancer to sensitize doxorubicin (Dox)-mediated apoptotic signaling, and this sensitization can be achieved by reducing c-FLIP expression via JNK activation and enhancing intracellular Dox accumulation via the inhibition of NF-κB activity. These findings suggest that HE in combination with Dox serves as an effective tool for treating drug-resistant human hepatocellular carcinoma.

© 2010 Elsevier Ireland Ltd. All rights reserved.

1. Introduction

Hepatocellular carcinoma (HCC) is the fifth most common cancer and the fourth leading cause of cancer-related mortality worldwide, with the highest incidence in Asia [1]. Despite surgical management and application of non-surgical therapeutic modalities, the incidence of HCC continues to rise [2]. Doxorubicin (Dox) is an effective chemotherapeutic drug for the treatment of patients with HCCs [3]. However, Dox has been shown to cause high systemic toxicity to normal tissues, as well as immunosuppression, secondary cardiomyopathy, and development of drug resistance during cancer therapy, thereby limiting successful outcomes of cancer chemotherapy [4,5]. Accordingly, several new therapeutic strategies are being developed to treat cancer. Recently, combination (rather than single-agent) chemotherapy has been found to be a superior treatment strategy that offers the potential for lowering the dose of chemotherapeutic drugs to reduce side effects. A related strategy is to evaluate the mechanisms of natural medicines [6,7]. Many studies have demonstrated that polysaccharides from basidiomycetes mushrooms had highly beneficial therapeutic effects including: (1) preventing oncogenesis after administration of peroral medications developed from these mushrooms or their extracts, (2) direct anti-tumor activity against various tumors, (3) synergistic anti-tumor activity in combination with chemotherapy, and (4) preventive effects on tumor metastasis [8–11]. In addition, unlike existing cancer chemotherapeutic drugs, mushroom-derived polysaccharides are known to have no toxic side effects. Most of the related clinical evidence was determined for the commercial polysaccharides lentinan (*Lentinus edodes*), krestin (*Coriolus versicolor*), and schizophyllan (*Schizophyllum commune*) [12]. Data on polysaccharides isolated from *Hericium erinaceus* (HE) is highly impressive, and these polysaccharides show promise for use as cancer therapeutics. Studies have shown that HE extracts strongly suppress the growth of various tumors *in vitro* and *in vivo* [13,14]. However, the molecular signaling involved in HE-mediated anti-tumor activity has never been investigated in HepG2 human hepatocellular carcinoma cells.

* Corresponding author. Address: College of Engineering, Department of Bioengineering and Technology, Kangwon National University, 192-1, Hyoja-2-dong, Chuncheon-si, Gangwon-do 200-701, Republic of Korea. Tel.: +82 33 250 6275; fax: +82 33 243 6350.
E-mail address: ekhong@kangwon.ac.kr (E.K. Hong).

0304-3835/$ - see front matter © 2010 Elsevier Ireland Ltd. All rights reserved.
doi:10.1016/j.canlet.2010.05.006

Apoptosis, or programmed cell death, is an important physiological process of cell death and occurs during tissue remodeling, immune regulation, and tumor regression. Most of the chemotherapeutic agents kill cancer cells by inducing apoptotic death pathways [15]. Cells undergoing apoptosis show a sequence of cardinal morphological features, including membrane blebbing, cellular shrinkage, and condensation of chromatin [16]. The two primary apoptotic pathways are the extrinsic death receptor-mediated pathway and the intrinsic mitochondria-mediated pathway. These two pathways, when stimulated, lead to the release of cytochrome c from the mitochondria and to the activation of the death signal [17]. Apoptotic signaling and execution through these two pathways depends on caspases, or aspartate-specific cysteine proteases, which are the key effector molecules in the apoptotic process [18]. It has been suggested that Dox, an anticancer agent, elicits a caspase cascade, leading to apoptosis [19,20]. A member of the mitogen-activated protein kinase (MAPK) family, c-Jun NH_2-terminal kinase (JNK), is rapidly phosphorylated and subsequently activated by a diverse spectrum of different cell stimuli. The JNK pathway has been shown to be closely linked to apoptosis by activation of the mitochondria-mediated apoptosis pathway and to regulate the degradation of cellular Fas-associated death domain (FADD) interleukin-1 β-converting enzyme (FLICE)-like inhibitory protein (c-FLIP), which is an important caspase cascade inhibitor [21–23].

To improve chemotherapeutic treatment, much effort has been made to identify chemosensitisers; that is, agents that are able to overcome multi-drug resistance (MDR) [24]. Interestingly, apart from MDR, nuclear factor κB (NF-κB) has been shown to be involved in chemoresistance in different cell types, suggesting a possible role of this transcription factor in the chemosensitising effects of agents [25].

Although there are many therapeutic strategies to treat cancer, including chemotherapy, high systemic cytotoxic toxicity and drug resistance have limited the successful outcomes in most cases. Recently, TNF-related apoptosis-inducing ligand (TRAIL) and non-steroidal anti-inflammatory drugs (NSAIDs) have been found to sensitize Dox-induced apoptosis in various types of tumors [26–28]. In addition, the combination treatment of Dox and natural products dramatically augments the therapeutic effects against breast cancer, prostate cancer, and hepatocellular carcinoma [29,30].

In this study, we investigated the ability of HE to synergize the inhibitory action of Dox in human hepatocellular carcinoma HepG2 cell lines. We also evaluated the molecular mechanisms underlying this synergistic effect and sought evidence of synergy between Dox and HE.

2. Materials and methods

2.1. Materials

The human hepatocellular carcinoma cell line, HepG2 (ATCC HB-8065), was obtained from Korean Cell Line Bank (Seoul, Korea). Fetal bovine serum, penicillin G, streptomycin, and RPMI 1640 were obtained from GIBCO (Grand Island, NY, USA). Dox, acridine orange, and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) were purchased from Sigma Chemical Co. (St. Louis, MO, USA). Cell-permeable inhibitor peptide (NF-κB SN50), Bay 11-7082, and SP600125 were obtained from Calbiochem (La Jolla, CA, USA). General caspase inhibitor z-VAD-fmk was purchased from BD Bioscience (San Diego, CA, USA). All other chemicals were of Sigma grade. Antibodies to caspase 3, p53, phospho-p53, JNK, phospho-JNK, NF-κB p105/p50, NF-κB p65, and horseradish peroxidase (HRP)-linked anti-rabbit IgG were obtained from Cell Signaling (Beverly, MA, USA). Antibodies to cytochrome c, Bax, Bcl-2, c-FLIP, β-actin, and HRP-linked goat anti-mouse IgG were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA).

2.2. Mushroom extracts and the chemotherapeutic drug

The crude water-soluble polysaccharide, which was obtained from the fruiting body of HE by hot water extraction and ethanol precipitation, was fractionated by DEAE-cellulose and Sepharose CL-6B column chromatography, as previously reported [31]. The purified components of HE primarily consisted of polysaccharide, which was a low-molecular-mass (13 kDa) 1,3-branched-β-1,6-glucan with a triple helix conformation. HE contained a level of endotoxin below the detection limits (0.0015 EU/ml) as assessed by an E-TOXATE kit (Sigma, St. Louis, MO, USA) [32].

Dox was dissolved in 1× phosphate–buffered saline (PBS) for injection at a concentration of 5 mg/ml and stored in aliquots at 4 °C. Immediately before each experiment, serial dilutions of this drug were performed in culture media to obtain the required final concentrations.

2.3. Cell culture

HepG2 cells were cultured in RPMI 1640 medium supplemented with 10% heat inactivated fetal bovine serum, 100 U/ml penicillin, and 100 μg/ml streptomycin. Cells were maintained at 37 °C in a 5% CO_2 incubator, and the media were changed twice weekly.

2.4. In vitro cytotoxic assay; colorimetric MTT assay

The cytotoxicity of HE and Dox on the viability of HepG2 cells was measured using the MTT assay, which relies on the ability of viable cells to metabolically reduce the tetrazolium salt MTT to a purple formazan product, which can be quantified colorimetrically [33]. Briefly, cell suspensions (2×10^5 cells/ml) were seeded in 96-well plates and incubated at 37 °C to allow for cell attachment. After 48 h, the cells were treated with serum-free medium containing the indicated additives. The plates were incubated for 1 h (d 0) and 72 h (d 3) at 37 °C. At the end of each exposure time, 50 μl of the MTT stock solution (5 mg/ml) in serum-free medium were added to each well to reach a total reaction volume of 200 μl. After incubation for 2 h at 37 °C, the supernatants were aspirated. The formazan crystals in each well were dissolved in 150 μl dimethylsulfoxide, and the A_{540} was read on a scanning multiwell spectrophotom-

eter. The relative percentage of survival was calculated by dividing the absorbance of treated cells by that of the control in each experiment. Percent cytotoxicity was calculated using the following formula: Percent cytotoxicity = [1 − (absorbance of experimental wells/absorbance of control wells)] × 100%.

2.5. Cell morphology assessment

Cells (2.5×10^5) were subcultured for 48 h on coverslips in six-well culture plates. After combination treatment of HE (100 μg/ml) with Dox (1 μg/ml) for 72 h, cell cytological morphology changes were observed using an Olympus optical microscope. Cells were fixed with 3.7% paraformaldehyde for 20 min, washed with PBS, and stained with acridine orange (5 μg/ml) for 5 min. After washing with PBS, cells were mounted under glass coverslips with Vectashield (Brunschwig, Amsterdam, The Netherlands) and observed. Photographs were taken under a fluorescence microscope.

2.6. Annexin V/propidium iodide staining

Apoptosis-mediated cell death of tumor cells was examined using a FITC-labeled annexin V/propidium iodide (PI) apoptosis detection kit (Molecular Probes, Eugene, OR, USA) according to the manufacturer's instructions. Briefly, Cells were harvested by trypsinization, washed with PBS, and centrifuged to collect the cell pellet. The number of cells was adjusted to 1×10^6 cell/ml. Cells were then resuspended in binding buffer (10 mM HEPES, 140 mM NaCl, 2.5 mM $CaCl_2$, pH 7.4), and stained with FITC-labeled annexin V and PI at room temperature for 15 min in the dark. Flow cytometric analysis was performed using a FACSCalibur flow cytometer (Becton Dickinson, Mountain View, CA, USA) within 1 h after supravital staining. FITC-labeled annexin V was analyzed using excitation and emission setting of 488 nm and 535 nm. PI was analyzed using excitation and emission setting of 488 nm and 575 nm. For each flow cytometer run, 10,000 cells were required. The percentages of cells were calculated by Cell Quest software (Becton Dickinson). The cells in the early stages of apoptosis were annexin V positive and PI negative; however, the cells in the late stages of apoptosis were both annexin V and PI positive. The apoptotic index (%) was calculated as the sum of early and late apoptotic cells divided for the total number of events.

2.7. Preparation of subcellular fractionation

The mitochondrial fraction was prepared using a mitochondria isolation kit (Sigma) according to the manufacturer's instructions. Briefly, after various treatments, HepG2 cells were harvested and resuspended in 0.65–2 ml of lysis buffer per $2–5 \times 10^7$ cells. The homogenate was incubated on ice for 5 min, two volumes of 1× extraction buffer were added, and the solution was centrifuged at 600*g* for 10 min at 4 °C. After centrifugation, the supernatant was transferred to fresh 1.5 ml tubes and centrifuged at 11,000*g* for 10 min at 4 °C. The supernatant was removed, and the pellet was suspended in a CelLytic M cell lysis reagent with protease inhibitor cocktail (1:100 [v/v]).

Nuclear extracts were prepared by lysing nuclei in a high salt buffer supplemented with protease and phosphatase inhibitors using a nuclear extraction kit (Panomics Inc., Fremont, CA, USA) according to the manufacturer's protocol. Protein concentrations were quantified by the Bio-Rad protein assay (Bio-Rad Laboratories, Hercules, CA, USA).

2.8. Western blot analysis

After treatment, cells were washed in 1× PBS and lysed in lysis buffer (10 mM Tris–HCl (pH 7.5), 10 mM NaH_2PO_4/$NaHPO_4$ (pH 7.5), 130 mM NaCl, 1% Triton X-100, 10 mM NaPPi, 1 mM phenylmethylsulphonyl fluoride, 2 μg/ml pepstatin A) for 30 min on ice. Lysates were centrifuged at 12,000*g* for 20 min at 4 °C. The supernatant was collected, and the protein content of the supernatant was measured using a Bio-Rad protein assay kit before analysis. The cytosolic, mitochondrial, or nuclear protein samples were loaded at 10 μg of protein/lane, separated by sodium dodecyl sulfate–polyacrylamide gel electrophoresis in 10–15% gel, and transferred to polyvinylidene difluoride membranes (Immun-Blot PVDF membrane, 0.2 μm; Bio-Rad). Membranes were blocked with 5% nonfat powdered milk in 1× tris buffered saline (TBS) containing 0.1% Tween 20 for 1 h, and they were incubated with primary antibodies at 4 °C overnight. Finally, the membranes were treated with horseradish peroxidase-coupled secondary antibodies for 1 h at 4 °C. The membranes were washed with T-TBS after each antibody binding reaction. Detection of each protein was performed using an enhanced chemiluminescence kit (Millipore Co., Billerica, MA, USA).

2.9. Assay for caspase 3 activities

After various treatments, HepG2 cells were lysed in 500 μl lysis buffer consisting of 10 mM Tris–HCl (pH 7.5), 10 mM NaH_2PO_4/$NaHPO_4$ (pH 7.5), 130 mM NaCl, 1% Triton X-100, and 10 mM NaPPi. Caspase 3 activity in the lysates was evaluated using a caspase assay kit (BD Bioscience) according to the manufacturer's instructions, with some modifications. For each sample, 50 μl of the cell lysate, 5 μl of Ac-DEVD-AMC (1 mg/ml; a synthetic fluorogenic substrate), and 50 μl of reaction buffer (20 mM HEPES at pH 7.5, 10% glycerol, and 2 mM DTT) were transferred to a 96-well plate, mixed, and incubated at 37 °C for 1.5 h. Caspase 3 cleaves Ac-DEVD-AMC to release the fluorescent AMC. The release of AMC in the cell lysates was measured using the Fluoroskan Ascent FL with Ascent software version 2.6 (Thermo scientific, Langenselbold, Germany) using an excitation filter with a wavelength of 360 nm and an emission filter with a wavelength of 460 nm [34].

2.10. Assay for caspase and JNK inhibitor activity

The cells were seeded at a density of 5×10^5 cells/well, and cultured for 48 h in RPMI 1640. The cells were preincubated with the indicated concentrations of z-VAD-fmk or SP600125 for 2 h, and they were treated with the

combination treatment of HE (100 μg/ml) and Dox (1 μg/ml) for 72 h. The viability of HepG2 cells was measured using the MTT assay.

2.11. Dox accumulation assay

Cells (2.5×10^5) were subcultured on coverslips in 6-well culture plates and incubated overnight, after which additives were added for 24 h. Subsequently, the culture medium was removed, and the cells were washed three times with PBS. Cells were fixed in 3.7% paraformaldehyde for 10 min at room temperature, washed three times with PBS, and mounted under glass coverslips with Vectashield. The cells were observed, and photographs for Dox fluorescence were taken under fluorescence microscopy.

2.12. Statistical analysis

Data were expressed as means ± standard error (SEM), and the results were taken from at least three independent experiments performed in triplicate. The data were analyzed by Student's *t*-test to evaluate significant differences. A level of $p < 0.05$ was regarded as statistically significant.

3. Results

3.1. HE and doxorubicin, at a low dose, synergise to induce apoptosis in human hepatocellular carcinoma cells

To investigate the anti-tumor effect of HE on hepatocellular carcinoma, we tested the effect of HE or Dox treatment on human hepatocellular carcinoma cells. HepG2 cells were either left untreated, or treated with different doses of HE or Dox for 72 h. After treatment with HE alone, a moderately dose-dependent growth-inhibitory effect on HepG2 cells was observed. A similar phenomenon was observed following Dox treatment (Fig. 1A). To evaluate whether or not the growth-inhibitory effect of HE or Dox treatment alone was associated with apoptosis, a double-staining method using FITC-labeled annexin V and PI was performed. During the early stage of apoptosis, cells display phosphatidylserine on their outer cell membranes, which is readily detectable by annexin V. During the later stages of apoptosis, as the plasma membrane becomes increasingly permeable, PI can move across the cell membrane to bind to cellular DNA. Double staining the cells with annexin V and PI allowed us to detect apoptotic cells by flow cytometry. Low doses of Dox (1 μg/ml) or HE (50 or 100 μg/ml) had a minor apoptotic effect on the cells. In comparison, treatment with high doses of Dox (5 μg/ml) caused more than 35% of the cells to undergo apoptosis (Fig. 1B). Since low doses of HE or Dox could not induce apoptosis, we tested the effect of the combination treatment of these two compounds, in the low dose range, on HepG2 cells that have demonstrated sensitivity to Dox treatment. Interestingly, the combination of low doses of HE and 1 μg/ml of Dox had a significantly ($p < 0.05$) greater dose-dependent inhibitory effect than that of HE or Dox alone at comparable concentrations (Fig. 2A). Moreover, the growth-inhibitory effect of the combination treatment on HepG2 cells was found to be time-dependent, because the inhibitory effect became gradually stronger with the passage of time after treatment; the most significant effect was observed at 72 h (Fig. 2B). To further confirm the cytotoxic synergy between HE and Dox, we performed a morphology study. Cells in dishes supplemented with 100 μg/ml of HE and 1 μg/ml of Dox became sparse, rounded, and detached from the dishes. We observed the normal morphology of drug-untreated HepG2 cells by staining cells with acridine orange. In this case, cells exhibited a shape that contained tri- or multi-angles with abundant cytoplasma and large oval nuclei with dispersed chromatin. Combination-treated HepG2 cells had the typical morphology of apoptotic cells, including cell shrinkage, chromatin condensation, and formation of apoptotic bodies (Fig. 2C). Normal rat hepatocytes were also used for testing the synergistic effect on the induction of apoptosis (Fig. 2D). Neither HE alone nor the combination treatment elicited apoptosis in the normal cells. Overall, the results indicated that low doses of HE and Dox synergistically enhanced Dox-induced apoptosis in HepG2 cells. We conducted further experiments to understand the molecular mechanism by which HE sensitizes Dox-induced apoptosis in HepG2 cells.

3.2. Involvement of caspases in the synergistic effect of HE and Dox

The caspase family plays an important role in many types of apoptosis, including Dox-induced apoptosis [35]. At high doses, Dox can induce apoptosis, possibly through caspase-mediated apoptotic signaling, including caspase 8 [19,20]. In a caspase cascade, caspase 8 and 9 act as initiators, whereas caspases 3 and 7 serve as effectors of apoptosis [15]. To study the mechanisms of the combination treatment of HE and Dox, we examined whether or not caspase was involved in the synergistic induction of apoptosis. Upon activation, caspases are cleaved into small, active forms [18]. Therefore, we analyzed changes in the levels of pro-caspase 3 by Western blot analysis. Treatment with low doses of HE or Dox did not induce caspase 3 cleavage. In comparison, pro-caspase 3 cleavage was remarkably induced in response to combination treatment with low doses of HE and Dox (Fig. 3A). To further determine the activation of caspase 3 by the combination treatment of HE and Dox, the activity of caspase 3 was analyzed using a commercial caspase 3 assay kit. The combination treatment significantly ($p < 0.05$) increased the activity of caspase 3 (Fig. 3B). To confirm whether or not the apoptosis induced by the combination treatment occurs via the caspase-dependent pathway, a general caspase inhibitor z-VAD-fmk was added to the combination treatment. Pretreatment of caspase inhibitor significantly ($p < 0.05$) reduced cytotoxicity by the combination treatment in a dose-dependent manner (Fig. 3C). Thus, these results indicated that low doses of HE and Dox could synergistically enhance caspase activity in HepG2 cells.

3.3. Combination treatment-mediated apoptosis via both extrinsic and intrinsic apoptotic pathways

Both the extrinsic death receptor-mediated and the intrinsic mitochondria-mediated apoptotic pathways play a critical role in apoptosis. The extrinsic apoptotic pathway comprises several protein members including the death receptors, the membrane-bound Fas ligand, the Fas complexes, the Fas-associated death domain, and caspase 8, which activate the rest of the downstream caspases leading to apoptosis. The intrinsic apoptotic pathway is dependent on the balance between Bcl-2 and Bax affecting the release of cytochrome c and second mitochondria-derived activator of caspase [15–17]. To examine whether the combination treatment of HE and Dox contributed to mitochondrial damage and to the activation of the downstream apoptosis cascade, the release of cytochrome c into the cytosolic fraction (an indicator of compromised mitochondrial integrity) was assessed using Western blot analysis. Cytochrome c was present in the cytosolic fraction and absent in the mitochondrial fraction isolated from the cells treated with the combination of HE and Dox (Fig. 4A). Bcl-2 family proteins serve as critical regulators of the mitochondria-mediated apoptosis pathway, functioning as either inhibitors (e.g., Bcl-2, Bcl-XL, Bcl-w and Mcl-2) or promoters (e.g., Bax, Bid, Bad and Bim) of cell death [36]. To further confirm that the mitochondrial apoptotic events were involved in the combination treatment-induced apoptosis, we analyzed changes in the levels of Bax and Bcl-2 by Western blot analysis. The combination treatment of HE and Dox increased expression of the pro-apoptotic protein Bax, but decreased expression of the anti-apoptotic protein Bcl-2 in HepG2 cells (Fig. 4B). c-FLIP, a dominant-negative inhibitor of caspase 8 activation, inhibits apoptosis induced by death receptors such as Fas or TRAIL receptors [37]. In addition, c-FLIP$_L$ can provide protection against intrinsic (Bax/mitochondria-dependent) apoptosis by inhibiting Bax activation [38]. We therefore examined c-FLIP protein expression by Western blot analysis in HepG2 cells. c-FLIP expression was dramatically suppressed by the combination treatment (Fig. 4C). The efficacy of Dox on cancer cells is in large part p53-dependent. As a result of the effect of p53 on the combination treatment-induced apoptosis using Western blot analysis, the phosphorylation of p53, a modulator of Bax, was affected in co-treated HepG2 cells (Fig. 4D). Overall, these results demonstrated that both the extrinsic and intrinsic apoptotic pathways were involved in the combination treatment-induced apoptosis in HepG2 cells.

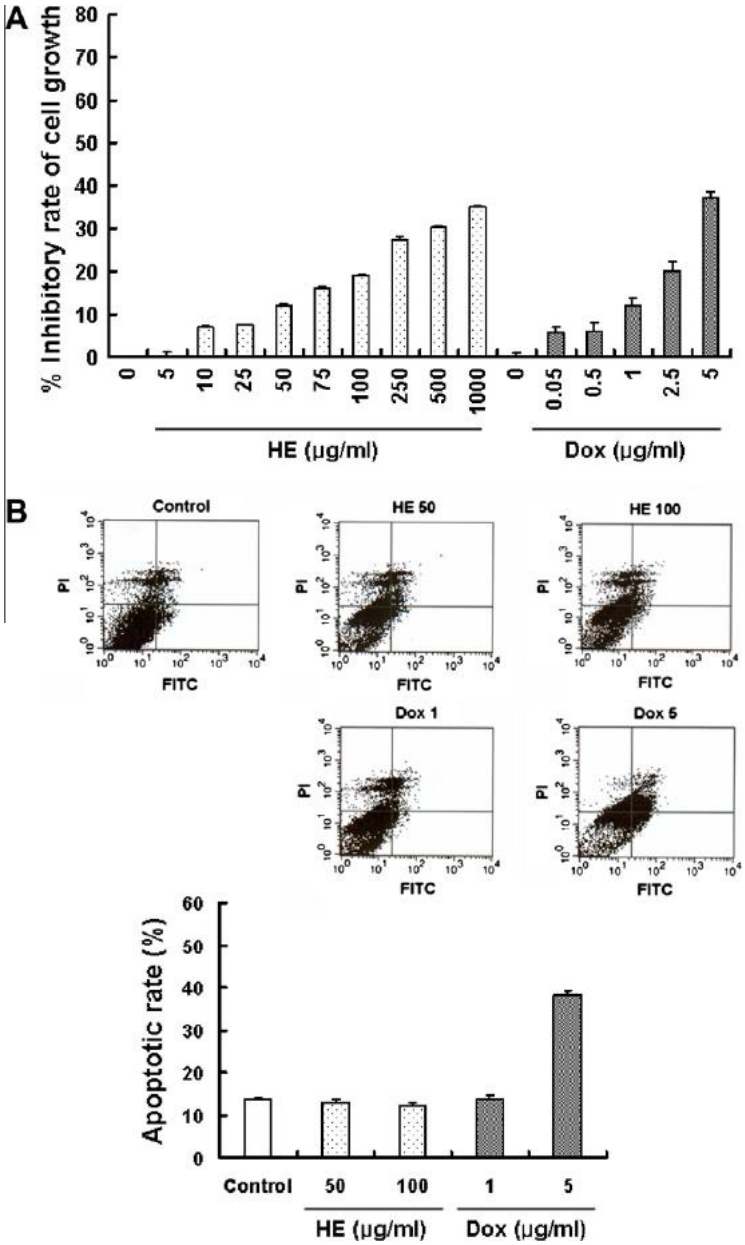

Fig. 1. Cell growth inhibition and apoptosis in HepG2 cells treated with various concentrations of HE and Dox. (A) The concentration-dependent growth-inhibitory effect of HE or Dox alone in HepG2 cells. Cells were incubated with various concentrations of HE or Dox for 72 h. Cell viability was measured by MTT reduction assay, as described in Section 2. Each value represents the mean ± SE of three independent experiments. (B) Percentages of apoptotic cells after various concentrations of HE or Dox treatment in HepG2 cells were measured by the annexin V/PI flow cytometry analysis, as described in Section 2. The *y*-axis plots the sum of early and late apoptotic cells as the mean ± SE of three independent experiments.

3.4. Activation of JNK is important for the induction of apoptosis in response to the combination treatment of HE and Dox

The JNK pathway has been shown to be closely linked to apoptosis by promoting the expression of pro-apoptotic factors, such as the Bcl-2 family proteins or decreasing the expression of anti-apoptotic factors. It has also been shown that activation of JNK during TNFα-induced apoptotic processes promotes the activity of caspases by proteasomal elimination of c-FLIP$_L$, an inhibitor of a caspase cascade [21]. To determine whether JNK activation was involved in the combination treatment-induced apoptosis, the phosphorylation status of JNK was determined by Western blot analysis. We observed that JNK phosphorylation was induced in HepG2 cells treated with the combination treatment of HE and Dox (Fig. 5A). To confirm the role of JNK activation in combination treatment-induced apoptosis, we combined JNK inhibitor SP600125 with the combination treatment of HE and Dox in the MTT assay and Western blot analysis. The JNK inhibitor significantly ($p < 0.05$) blocked the inhibitory rate of cell growth in a dose-dependent manner. Furthermore, the increased expression of c-FLIP by Western blot analysis suggested that JNK activity, at least in part, was involved in the reduction of c-FLIP expression in the cytotoxicity rendered by the combination treatment of HE and Dox (Fig. 5B). Recent studies have suggested that the ability of JNK to regulate apoptosis

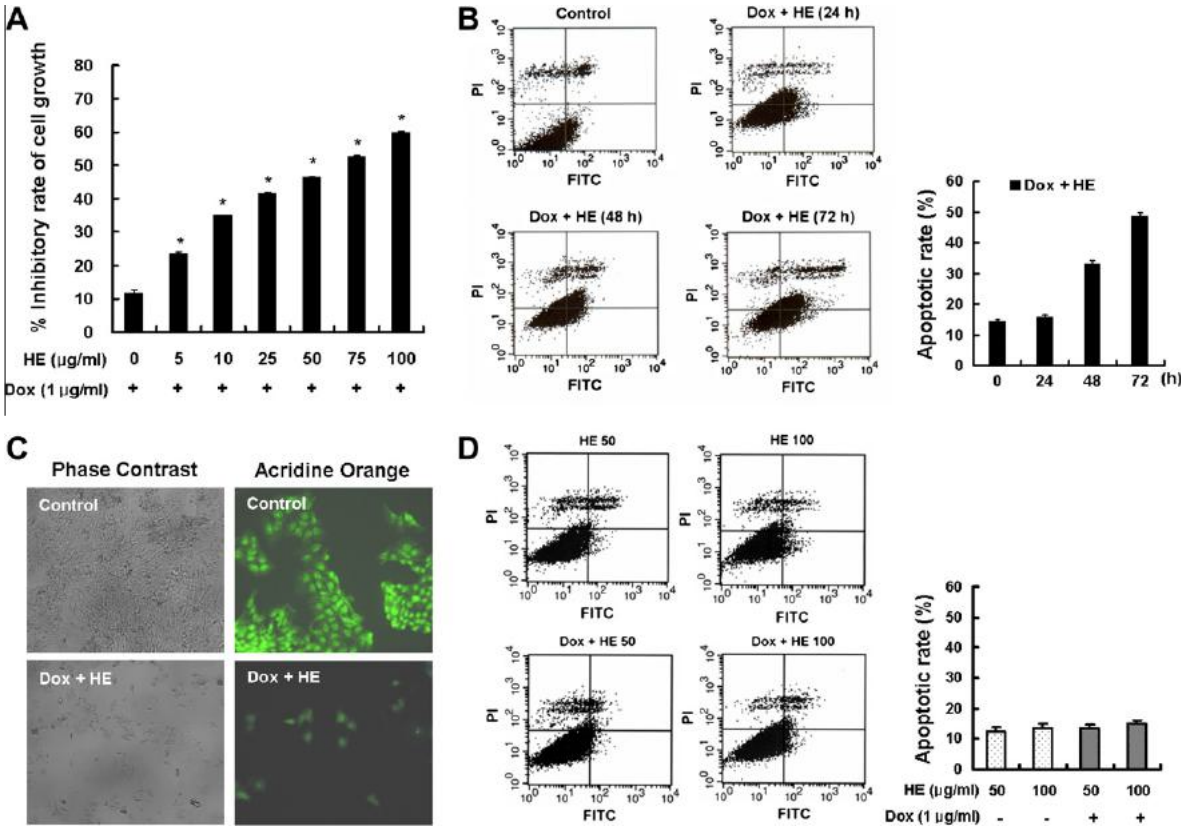

Fig. 2. Synergistic cell growth inhibition and apoptosis in HepG2 cells treated with HE in combination with Dox. (A) The concentration-dependent synergistic cell growth-inhibitory effect of various concentrations of HE in combination with Dox. The cells incubated with various concentrations of HE in combination with Dox (1 μg/ml) for 72 h. Cell viability was measured by the MTT reduction assay, as described in Section 2. Each value represents the mean ± SE of three independent experiments. Significant differences were compared with the control at $^{*}p < 0.05$ by Student's *t*-test. (B) Percentages of apoptotic cells after the combination treatment of HE (100 μg/ml) and Dox (1 μg/ml) in HepG2 cells at 0 h, 24 h, 48 h, and 72 h were measured by the annexin V/PI flow cytometry analysis, as described in Section 2. The *y*-axis plots the sum of early and late apoptotic cells as the mean ± SE of three independent experiments. (C) Morphological characteristics of HepG2 cells before and after the combination treatment. The HepG2 cells were treated with or without HE (100 μg/ml) and Dox (1 μg/ml) for 72 h and stained with acridine orange. The cell morphology was observed under optical microscope (magnification: 100×) and fluorescence microscope (magnification: 200×). (D) Examination of cytotoxic effect of HE and Dox on rate hepatocytes. Hepatocytes were treated with HE (50 and 100 μg/ml) or HE plus Dox (1 μg/ml) for 72 h. Percentages of apoptotic cells were measured by the annexin V/PI flow cytometry analysis, as described in Section 2. The *y*-axis plots the sum of early and late apoptotic cells as the mean ± SE of three independent experiments.

does not depend on p53 function [39]. To ascertain p53-dependent and -independent consequences of JNK activation, we combined the JNK inhibitor SP600125 with the combination treatment of HE and Dox in the MTT assay and Western blot analysis. The JNK inhibitor blocked the inhibitory rate of cell growth in a dose-dependent manner; however, it did not affect the phosphorylation of p53 using Western blot analysis, suggesting that the ability of JNK to regulate apoptosis did not depend on p53 function (Fig. 5B).

3.5. HE increased the intracellular accumulation and retention of Dox

Dox is DNA-damaging agent. DNA damage is caused by ATM and Chk2 kinases, which phosphorylate and stabilize p53, allowing it to function as a transcription factor inducing apoptosis [40]. To understand the enhancement of the phosphorylation of p53 in the combination treatment of HE and Dox, we investigated the effect of HE on intracellular accumulation of Dox. Dox is an auto-fluorescent compound that allows for the visualization of its intracellular presence by fluorescence microscopy. To investigate the effect of HE on intracellular accumulation of Dox, we examined the fluorescence of intracellular-accumulated Dox in HepG2 cells after incubation for 24 h in the absence or presence of HE (100 μg/ml). The intracellular fluorescence, which was primarily visible in the nuclei, was remarkably higher in cells that were co-treated with HE (Fig. 6A). To determine the effect of HE on the intracellular retention of Dox, HepG2 cells were incubated for 24 h with Dox (1 μg/ml) followed by incubation for an additional 24 h in a control medium or in a medium containing HE (100 μg/ml). Cells treated with HE contained more Dox than those that were incubated in the control medium during the second 24 h culture period (Fig. 6B). Therefore, these results indicated that HE enhanced Dox-induced cytotoxicity in HepG2 cells was due to an increase in intracellular Dox accumulation.

3.6. The combination treatment of HE and Dox suppress NF-κB translocation

NF-κB has been shown to be involved in chemoresistance in different cell types, suggesting a possible role of this transcription factor in the chemosensitising effect of agents [25,41]. To unravel the molecular mechanisms involved in Dox accumulation, we examined the effect of the combination treatment of HE and Dox on the translocation of NF-κB by Western blot analysis using an NF-κB subunit p65 antibody. Dox strongly induced the translocation of nuclear p65 into the nuclei; however, the combination treatment remarkably inhibited Dox-enhanced nuclear p65 translocation (Fig. 7).

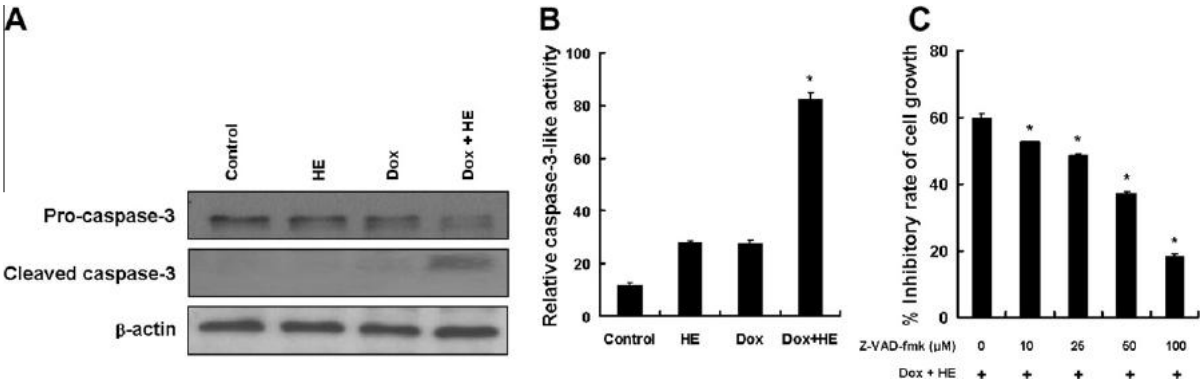

Fig. 3. Activation of caspase 3 following the combination treatment of HE and Dox. (A) After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, HepG2 cells were harvested, and lysates were prepared. The cleavage of pro-caspase 3 and existence of the activated form of caspase 3 were determined by Western blot analysis. Equal loading of total proteins in each sample was verified by β-actin expression. (B) Effect of the treatment of HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox on caspase 3-like activity in HepG2 cells. After treatment for 72 h, preparation of cell lysates and measurement of caspase 3-like activity was performed, as described in Section 2. Each value represents the mean ± SE of three independent experiments. Significant differences were compared with the control at $^{*}p < 0.05$ by Student's *t*-test. (C) Effects of the caspase inhibitor z-VAD-fmk on the synergistic cell growth inhibition in HepG2 cells treated with HE plus Dox. HepG2 cells were pre-incubated with the indicated concentrations of z-VAD-fmk for 2 h then treated with HE plus Dox for 72 h. Cell viability was measured by the MTT reduction assay, as described in Section 2. Each value represents the mean ± SE of three independent experiments. Significant differences were compared with the control at $^{*}p < 0.05$ by Student's *t*-test.

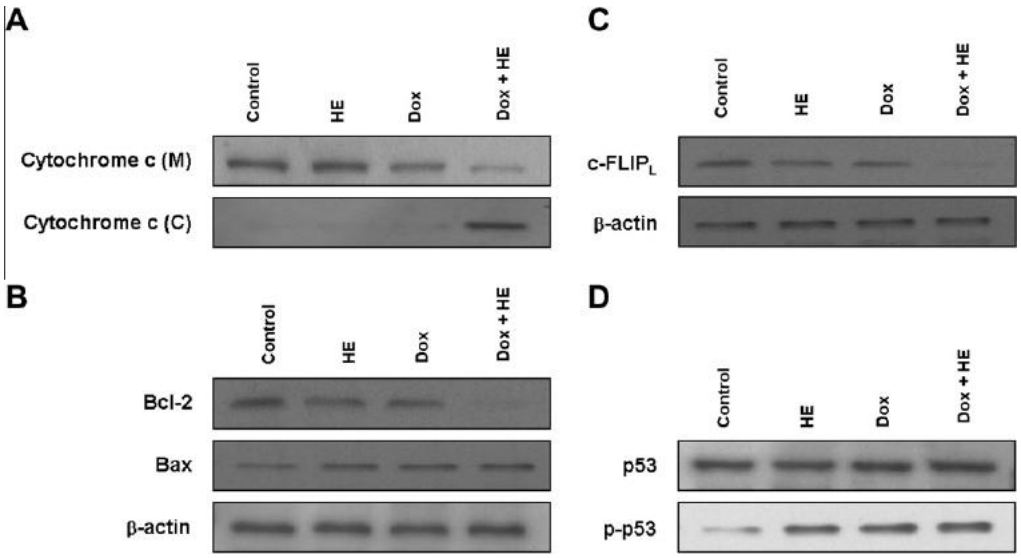

Fig. 4. Altered expression of apoptotic-related proteins in HepG2 cells treated with HE, Dox, or HE plus Dox. (A) The release of cytochrome c into the cytosol following treatment with HE, Dox, or HE plus Dox. After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, the mitochondrial (M) and cytosolic (C) fractions from the indicated additive-treated cells were isolated and analyzed for the expression of cytochrome c using Western blot analysis. (B) Expression level of Bcl-2 family proteins following treatment with HE, Dox, or HE plus Dox. After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, HepG2 cells were harvested, and lysates were prepared. The levels of Bcl-2 and Bax were assessed using Western blot analysis. Equal loading of total proteins in each sample was verified by β-actin expression. (C) c-FLIP$_L$ expression upon treatment with HE, Dox, or HE plus Dox. After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, HepG2 cells were harvested, and lysates were prepared. The level of c-FLIP$_L$ was detected using Western blot analysis. Equal loading of total proteins in each sample was verified by β-actin expression. (D) p53 expression and its phosphorylation status upon treatment with HE, Dox, or HE plus Dox. After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, HepG2 cells were harvested, and lysates were prepared. The levels of p53 and phospho-p53 were assessed using Western blot analysis.

3.7. The inhibition of NF-κB activation was involved in modulating intracellular Dox accumulation

To further investigate the role of NF-κB in the enhanced anti-tumor efficacy of Dox, we examined the effects of Dox in the absence or presence of NF-κB inhibitors, BAY 11-7082 (10 μM) or SN50 (25 μM), on HepG2 cell growth and intracellular Dox accumulation. The NF-κB inhibitors, BAY 11-7082 or SN50, had a greater time-dependent anti-proliferative effect than that of Dox alone, and they also increased intracellular Dox accumulation (Fig. 8A and B). Furthermore, Western blot analysis of another NF-κB subunit p50 in nuclear (N) and cytosolic (C) extracts from HepG2 cultured for 24 h in the treatment with Dox (alone or in combination with BAY 11-7082 or SN50) showed that BAY 11-7082 and SN50 inhibited Dox-enhanced nuclear p50 translocation (Fig. 8C). Thus, these results showed that the inhibition of NF-κB activation might be involved in modulating intracellular Dox accumulation.

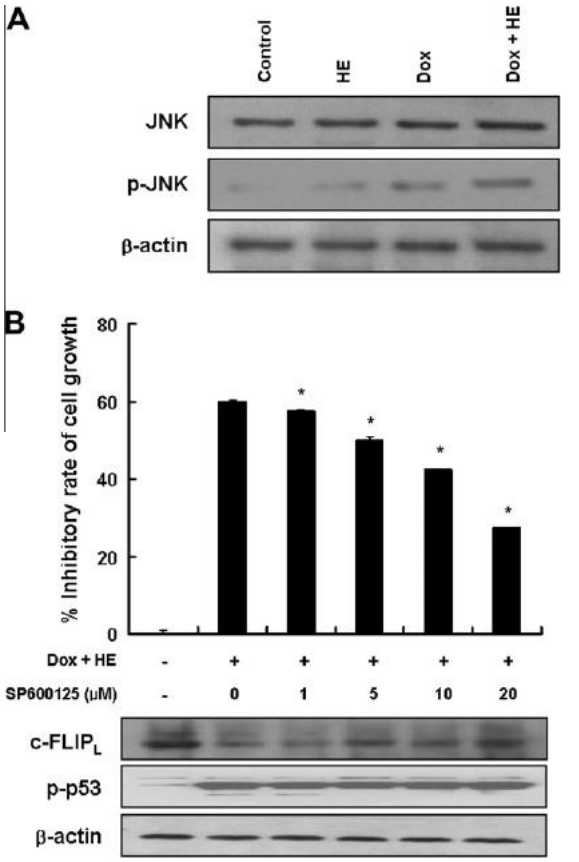

Fig. 5. JNK activation, c-FLIP$_L$ expression, and the phosphorylation status of p53 following the combination treatment of HE and Dox. (A) After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 72 h, HepG2 cells were harvested, and lysates were prepared. The levels of JNK and phospho-JNK were assessed using Western blot analysis. Equal loading of total proteins in each sample was verified by β-actin expression. (B) Effects of SP600125 on the synergistic cell growth inhibition, c-FLIP$_L$ expression, and the phosphorylation status of p53 in HepG2 cells treated with HE plus Dox. HepG2 cells were pre-incubated with the indicated concentrations of SP600125 for 2 h then treated with HE plus Dox for 72 h. Cell viability was measured by the MTT reduction assay, as described in Section 2. Each value represents the mean ± SE of three independent experiments. Significant differences were compared with the control at $^*p < 0.05$ by Student's *t*-test. HepG2 cells were pre-incubated with the indicated concentrations of SP600125 for 2 h then treated with HE plus Dox for 72 h. HepG2 cells were harvested, and lysates were prepared. The levels of c-FLIP$_L$ and phospho-p53 were assessed using Western blot analysis. Equal loading of total proteins in each sample was verified by β-actin expression.

4. Discussion

In this study, we showed that low doses of HE and Dox synergize to induce apoptosis in human hepatocellular carcinoma HepG2 cells. Caspase 3 was activated in the apoptotic process induced by the combination treatment of HE and Dox. The suppression of caspase activity by a general caspase inhibitor z-VAD-fmk confirmed that the promotion of apoptosis by combination treatment involved a caspase-dependent pathway. Moreover, reducing the expression level of c-FLIP through upregulation of JNK activity and reducing the ratio of Bcl-2/Bax protein expression through the phosphorylation of p53 and/or downregulation of c-FLIP$_L$ induced the release of cytochrome c from the mitochondria to cytosol. The addition of the JNK inhibitor SP600125 suppressed the apoptotic process induced by combination treatment with HE and Dox. Therefore, we have been able to show that both the extrinsic and intrinsic apoptotic pathways were involved in the combination treatment-induced apoptosis in HepG2 cells.

Cancer progression is partially affected by the development of cancer cell resistance to apoptosis induced by chemotherapeutic drugs. One feature that cytotoxic treatments of cancer have in common is their activation of the transcription factor NF-κB, which regulates cell survival [41,42]. NF-κB activation suppresses the apoptotic potential of chemotherapeutic drugs and contributes to resistance. In this approach, the ability to sensitize apoptotic signaling pathways implicated in enhancing chemosensitivity of chemotherapeutic drugs has important therapeutic implications [25]. In this study, HE increased the intracellular accumulation and retention of Dox. We also showed that, like HE, NF-κB inhibitors (BAY 11-7082 or SN50) increased the intracellular accumulation of Dox and augmented the Dox-induced cytotoxicity mediated via suppression of NF-κB activity. Therefore, the combination treatment of low doses of HE and Dox, which enhanced chemosensitivity by suppression of NF-κB translocation, would have similar implications for the treatment of patients with HCCs.

The two primary apoptotic pathways are the extrinsic death receptor-mediated pathway and the intrinsic mitochondria-mediated pathway [15–17]. In response to diverse stimuli, pro-apoptotic Bcl-2 family proteins such as Bax initiate the intrinsic apoptotic pathway by forming channels on assimilating into the mitochondria, thus increasing outer mitochondrial membrane permeability, and thereby facilitating the release of cytochrome c and other pro-apoptotic factors from the mitochondrial intermembrane space. Released cytochrome c forms an apoptosome complex with Apaf-1, which activates caspase 9, and in turn, its downstream caspase 3, resulting in the morphological features of apoptosis [18]. Dox is DNA-damaging agent. DNA damage is caused by ATM and Chk2 kinases, which phosphorylate and stabilize p53, allowing it to function as a transcription factor inducing apoptosis [40]. The efficacy of a low dose of Dox on cancer cells is in large part p53-dependent [19,20]. It was found that p53 mediated apoptosis by transcriptionally regulating many other genes, including Bax, Apaf-1, Fas, or killer/DR5 [40]. In addition, recent studies support a transcriptionally independent pathway involving localization of p53 to the mitochondria, followed by binding and inhibition of Bcl-xL and/or Bcl-2 or by activating Bax directly [43]. In agreement with this, the combination treatment of HE and Dox upregulated the phosphorylation of p53, increased the expression of the pro-apoptotic protein Bax, and decreased the expression of anti-apoptotic Bcl-2, increased the release of cytochrome c from the mitochondria to cytosol, and subsequently activated caspase 3. A universal caspase inhibitor blocked this synergistic cytotoxic effect on HepG2 cells, further suggesting the involvement of caspase activity in the cytotoxicity rendered by the combination treat-

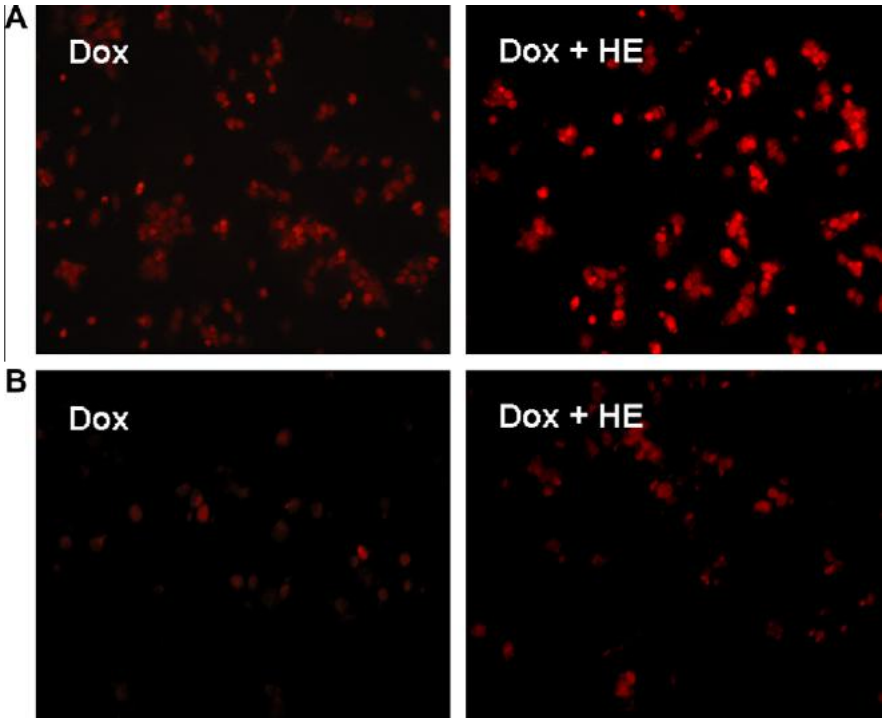

Fig. 6. Effect of HE on the intracellular accumulation and retention of Dox. (A) Intracellular Dox accumulation: fluorescent microscopic pictures of HepG2 cells that were incubated for 24 h in a medium containing Dox (1 μg/ml) in combination with HE (100 μg/ml). Magnification: 200×. (B) Intracellular Dox retention: Fluorescent microscopic pictures of HepG2 cells treated with Dox (1 μg/ml) for 24 h, followed by 24 h incubation in control medium or medium containing HE (100 μg/ml). Magnification: 200×.

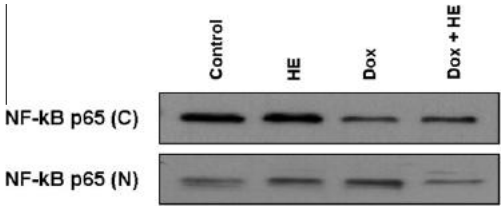

Fig. 7. Suppression of NF-κB subunit p65 translocation following combination treatment with HE and Dox. After treating with HE (100 μg/ml), Dox (1 μg/ml), or HE plus Dox for 24 h, the nuclear (N) and cytosolic (C) fractions from the indicated additive-treated cells were isolated and analyzed for the expression of p65 using Western blot analysis.

ment with HE and Dox. Therefore, these results support the involvement of the intrinsic mitochondria-mediated apoptotic pathway in combination treatment-induced apoptosis in HepG2 cells.

The extrinsic apoptotic pathway involves death receptors, which upregulate the downstream signaling cascade leading to the activation of caspase 8 [17]. The activated caspase 8 cleaves downstream caspases such as caspases 3, 6, and 7, as well as the pro-apoptotic protein Bid which initiates the apoptotic process. Cleaved Bid activates downstream pro-apoptotic proteins such as Bax which promote cytochrome c release from the mitochondria into the cytosol, thereby linking the two pathways [22]. The intrinsic and extrinsic apoptotic pathways converge to caspase 3, which cleaves the inhibitor of the caspase-activated deoxyribonuclease, and the caspase 3-activated deoxyribonuclease becomes active leading to nuclear apoptosis [15]. The cancer chemotherapeutic drug Dox has been shown to induce apoptosis by initiating a caspase cascade. Previous studies using specific caspase inhibitors have shown that Dox can activate caspases 8, 6, and 3 [19,20]. c-FLIP is an inhibitor of apoptosis downstream of the death receptors. Following death receptor engagement, c-FLIP inhibits caspase 8 binding to FADD and prevents death inducing signaling complex (DISC) formation and apoptosis [37]. Previous studies have shown that JNK activation in TNFα-induced apoptotic processes promotes the proteasomal elimination of c-$FLIP_L$, an inhibitor of caspase 8 [21]. In line with these previous studies, our data showed that Dox also was able to synergize with HE at low doses for the induction of apoptosis [26,29,44]. In this apoptotic process, the expression level of caspase 8 inhibitor c-FLIP through upregulation of JNK activation decreased; subsequently, caspase 3 was activated [45]. These results support the involvement of the extrinsic death receptor-mediated apoptotic pathway in combination treatment-induced apoptosis in HepG2 cells.

The JNK signaling pathway, though not a direct apoptosis executor, is one of the most important apoptosis-signaling pathways. JNK modulates the initiation of apoptosis by promoting the process of pro-apoptotic factors, such as the Bcl-2 family proteins or the decaying of anti-apoptotic factors [22,23]. In our study, the combination treatment of HE and Dox elicited JNK phosphorylation, which was responsi-

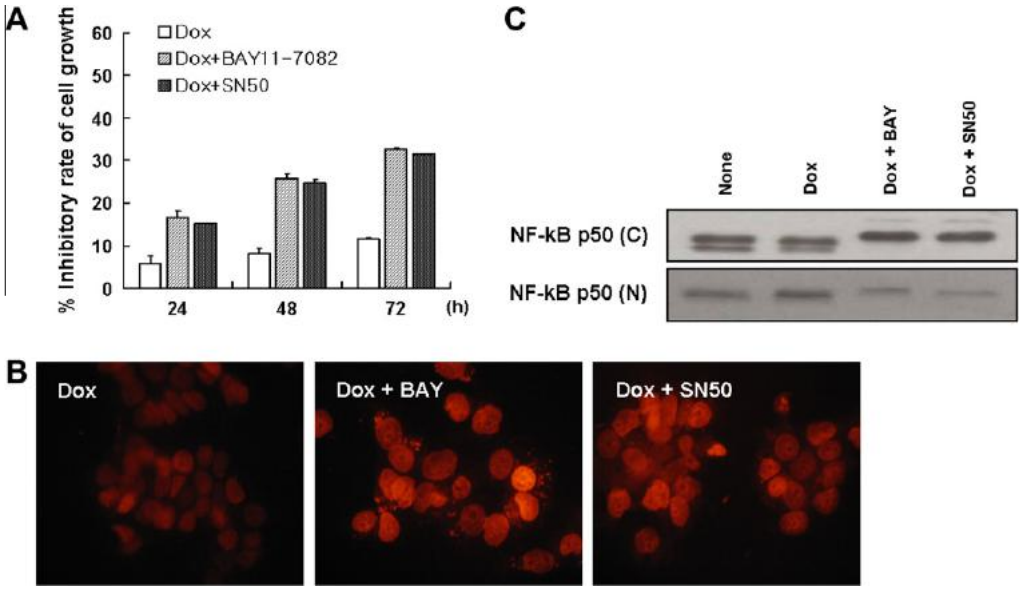

Fig. 8. Effect of Dox, alone or in combination with NF-κB inhibitor on HepG2 cell growth and effect of NF-κB inhibitor on intracellular Dox accumulation. (A) A time-dependent synergistic cell growth-inhibitory effect of BAY 11-7082 or SN50 in combination with Dox. HepG2 cells were pre-incubated with BAY 11-7082 (10 μM) or SN50 (25 μM) for 2 h then treated with Dox (1 μg/ml) for 24 h, 48 h, or 72 h. Cell viability was measured by the MTT reduction assay, as described in Section 2. Each value represents the mean ± SE of three independent experiments. (B) Fluorescent microscopic pictures of HepG2 cells that were incubated for 24 h in a medium containing Dox (1 μg/ml) or in combination with BAY 11-7082 (10 μM) or SN50 (25 μM). Magnification: 400×. (C) Suppression of the NF-κB subunit p50 translocation following combination treatment with NF-κB inhibitor and Dox. HepG2 cells were pre-incubated with BAY 11-7082 (10 μM) or SN50 (25 μM) for 2 h then treated with Dox (1 μg/ml) for 24 h. The nuclear (N) and cytosolic (C) fractions from the indicated additive-treated cells were isolated and analyzed for the expression of p50 using Western blot analysis.

ble for the reduction of c-FLIP$_L$ expression in HepG2 cells, suggesting the involvement of JNK activity in the cytotoxicity rendered by combination treatment with HE and Dox. Recently, the effect of SP600125 on chemotherapy-induced apoptosis was shown to be more selective for paclitaxel rather than Dox, suggesting that the ability of JNK to regulate apoptosis does not depend on p53 function [39]. In our experimental setting, addition of the JNK inhibitor SP600125 suppressed the apoptotic process induced by the combination treatment of HE and Dox, increased the expression of c-FLIP$_L$, and did not affect the phosphorylation status of p53. Therefore, JNK, by promoting the degradation process, suppressed c-FLIP$_L$ and might sensitize the hepatocellular carcinoma cells to apoptosis [46].

One feature that cytotoxic treatments of cancer have in common is their activation of the transcription factor NF-κB, which suppresses the apoptotic potential of chemotherapeutic drugs and contributes to resistance [40,41]. Inhibition of NF-κB translocation has been implicated in enhanced chemosensitivity of chemotherapeutic drugs [25]. Using Western blot analysis we found that combination treatment with HE and Dox could inhibit NF-κB nuclear translocation, whereas Dox increased it. To further confirm the role of NF-κB in the enhanced cytotoxic efficacy of Dox, we combined NF-κB inhibitors BAY 11-7082 or SN50 with Dox in the MTT and fluorescent Dox accumulation assays. NF-κB inhibitors inhibited Dox-enhanced p50 nuclear translocation in Western blot analysis, increased the cytotoxic effect of Dox in the MTT assay in a time-dependent manner, and increased the intracellular accumulation of Dox. The molecular mechanism linking inhibition of NF-κB activity with enhanced Dox accumulation may be associated with the membrane-related actions of Dox and the decrease high basal expression of MDR gene and transglutaminase [47–49].

In summary, chemotherapeutic resistance to drugs is a major obstacle to the successful treatment of human hepatocellular carcinoma. We have demonstrated that HE acts as an enhancer to sensitize Dox-mediated apoptotic signaling by reducing c-FLIP expression via JNK activation and enhancing intracellular Dox accumulation via the inhibition of NF-κB activity. Therefore, HE, when combined with low doses of Dox, has the potential to provide more efficient therapeutic effects against drug-resistant human hepatocellular carcinoma.

Conflicts of interest

None declared.

References

[1] G. Falkson, J.M. MacIntyre, C.G. Moertel, Primary liver cancer, Cancer 54 (1994) 977–980.
[2] J.M. Llovet, Updated treatment approach to hepatocellular carcinoma, J. Gastroenterol. 40 (2005) 225–235.
[3] H.S. Hochster, M.D. Green, J. Speyer, E. Fazzini, R. Blum, F.M. Muggia, 4'Epidoxorubicin (epirubicin): activity in hepatocellular carcinoma, J. Clin. Oncol. 3 (1985) 1535–1540.
[4] D.D. Von Hoff, M.W. Layard, P. Basa, H.L. Davis, A.L. Von Hoff Jr., M. Rozencweig, F.M. Muggia, Risk factors for doxorubicin-induced congestive heart failure, Ann. Internai Med. 91 (1979) 710–717.

[5] E. Saltiel, W. McGuire, Doxorubicin (adriamycin) cardiomyopathy, Western J. Med. 139 (1983) 332–341.
[6] R.G. Mehta, J.M. Pezzuto, Discovery of cancer preventive agents from natural products: from plants to prevention, Curr. Oncol. Rep. 4 (2002) 478–486.
[7] P.M. Kidd, The use of mushroom glucans and proteoglycans in cancer treatment, Alternative Med. Rev. 5 (2000) 4–27.
[8] H. Kobayashi, R. Yoshida, Y. Kanada, Y. Fukuda, T. Yagyu, K. Inagaki, T. Kondo, N. Kurita, M. Suzuki, N. Kanayama, T. Terao, Suppressing effects of daily oral supplementation of beta-glucan extracted from *Agaricus blazei* Murill on spontaneous and peritoneal disseminated metastasis in mouse model, J. Cancer Res. Clin. Oncol. 131 (2005) 527–538.
[9] G. Chihara, Y. Maeda, J. Hamuro, T. Sasaki, F. Fukuoka, Inhibition of mouse sarcoma 180 by polysaccharides from *Lentinus edodes* (Berk.) sing, Nature 222 (1969) 687–688.
[10] J.M. Wan, W.H. Sit, J.C. Louie, Polysaccharopeptide enhances the anticancer activity of doxorubicin and etoposide on human breast cancer cells ZR-75-30, Int. J. Oncol. 32 (2008) 689–699.
[11] T.B. Ng, H.X. Wang, Pharmacological actions of *Cordyceps*, a prized folk medicine, J. Pharm. Pharmacol. 57 (2005) 1509–1520.
[12] S.P. Wasser, Medicinal mushrooms as a source of antitumor and immunomodulating polysaccharides, Appl. Microbiol. Biotechnol. 60 (2002) 258–274.
[13] J.C. Wang, S.H. Hu, C.H. Su, T.M. Lee, Antitumor and immunoenhancing activities of polysaccharides from culture broth of *Hericium* spp., Kaohsiung J. Med. Sci. 17 (2001) 461–467.
[14] T. Mizuno, T. Wasa, H. Ito, C. Suzuki, N. Ukai, Antitumor-active polysaccharides isolated from the fruiting body of *Hericium erinaceum*, an edible and medicinal mushroom called yamabushitake or houtou, Biosci. Biotechnol. Biochem. 56 (1992) 347–348.
[15] I.M. Ghobrial, T.E. Witzig, A.A. Adjei, Targeting apoptosis pathways in cancer therapy, CA, Cancer J. Clin. 55 (2005) 178–194.
[16] A. Saraste, K. Pulkki, Morphologic and biochemical hallmarks of apoptosis, Cardiovasc. Res. 45 (2000) 528–537.
[17] K. Schulze-Osthoff, D. Ferrari, M. Los, S. Wesselborg, M.E. Peter, Apoptosis signaling by death receptors, Eur. J. Biochem. 254 (1998) 439–459.
[18] G.M. Cohen, Caspases: the executioners of apoptosis, Biochem. J. 326 (1997) 1–16.
[19] Y.A. Hannun, Apoptosis and the dilemma of cancer chemotherapy, Blood 89 (1997) 1845–1853.
[20] M. Los, I. Herr, G. Friesen, S. Fulda, K. Schulze-Osthoff, S. Lipkowitz, Cross-resistance of CD95 and drug-induced apoptosis as a consequence of deficient activation of caspases (ICE/Ced-3 proteases), Blood 90 (1997) 3118–3129.
[21] L. Chang, H. Kamata, G. Solinas, J. Luo, S. Maeda, K. Venuprasad, Y. Liu, M. Karin, The E3 ubiquitin ligase Itch couples JNK activation to TNFα-induced cell death by inducing c-FLIP$_L$ turnover, Cell 124 (2006) 601–613.
[22] V. Gogvadze, S. Orrenius, Mitochondrial regulation of apoptotic cell death, Chem. Biol. Interact. 163 (2006) 4–14.
[23] K. Zu, L. Hawthorn, C. Ip, Up-regulation of c-Jun-NH2-kinase pathway contributes to the induction of mitochondria-mediated apoptosis by alpha-tocopheryl succinate in human prostate cancer cells, Mol. Cancer Ther. 4 (2005) 43–50.
[24] G.D. Kruh, L.J. Goldstein, Doxorubicin and multidrug resistance, Curr. Opin. Oncol. 5 (1993) 1029–1034.
[25] C. Nakanishi, M. Toi, Nuclear factor-κB inhibitors as sensitizers to anticancer drugs, Nat. Rev. Cancer 5 (2005) 297–309.
[26] X. Wu, Y. Kakehi, Y. Mizutani, T. Kamoto, H. Kinoshita, Y. Isogawa, T. Terachi, O. Ogawa, Doxorubicin enhances TRAIL-induced apoptosis in prostate cancer, Int. J. Oncol. 20 (2002) 949–954.
[27] Y.Y. Pan, S.P. Xu, X.Y. Jia, H.Q. Xu, Y. Zhang, L.X. Rui, W. Wei, Combination of cyclooxygenase-2 inhibitor and doxorubicin increases the growth inhibition and apoptosis in human hepatocellular carcinoma cells, Exp. Oncol. 29 (2007) 23–29.
[28] J. van Wijngaarden, E. van Beek, G. van Rossum, C. van der Bent, K. Hoekman, G. van der Pluijm, M.A. van der Pol, H.J. Broxterman, V.W. van Hinsbergh, C.W. Lowik, Celecoxib enhances doxorubicin-induced cytotoxicity in MDA-MB231 cells by NF-κB-mediated increase of intracellular doxorubicin accumulation, Eur. J. Cancer 43 (2007) 433–442.
[29] L. Collins, T. Zhu, J. Guo, Z.J. Xiao, C.Y. Chen, *Phellinus linteus* senstises apoptosis induced by doxorubicin in prostate cancer, Brit. J. Cancer 95 (2006) 282–288.
[30] K. Pinmai, S. Chunlaratthanabhorn, C. Ngamkitidechakui, N. Soonthornchareon, C. Hahnvajanawong, Synergistic growth inhibitory effects of *Phyllanthus emblica* and *Terminalia bellerica* extracts with conventional cytotoxic agents: doxorubicin and cisplatin against human hepatocellular carcinoma and lung cancer cells, World J. Gastroenterol. 14 (2008) 1491–1497.
[31] J.S. Lee, J.S. Kwon, J.S. Yun, J.W. Pahk, W.C. Shin, S.Y. Lee, E.K. Hong, Structural characterization of immunostimulating polysaccharide from cultured mycelia of *Cordyceps militaris*, Carbohyd. Polym. 80 (2010) 1011–1017.
[32] J.S. Lee, K.M. Min, J.Y. Cho, E.K. Hong, Study of macrophage activation and structural characteristics of purified polysaccharides form the fruiting body of *Hericium erinaceus*, J. Microbiol. Biotechnol. 19 (2009) 951–959.
[33] J.S. Lee, J.Y. Cho, E.K. Hong, Study on macrophage activation and structural characteristics of purified polysaccharides from the liquid culture broth of *Hericium erinaceus*, Carbohyd. Polym. 78 (2009) 162–168.
[34] G. Feng, N. Kaplowitz, Colchicine protects mice from the lethal effects of an agonistic anti-Fas antibody, J. Clin. Invest. 105 (2000) 329–339.
[35] N.A. Thornberry, Y. Lazebnik, Caspases: enemies within, Science 281 (1998) 1312–1316.
[36] A. Gross, J.M. McDonnell, S.J. Korsmeyer, Bcl-2 family members and the mitochondria in apoptosis, Genes Dev. 13 (1999) 1899–1911.
[37] T.W. Day, S. Huang, A.R. Safa, C-FLIP knockdown induces ligand-independent DR5-, FADD-, caspase-8-, and caspase-9-dependent apoptosis in breast cancer cells, Biochem. Pharmacol. 76 (2008) 1694–1704.
[38] X. Wang, Y. Wang, H.P. Kim, A.M. Choi, S.W. Ryter, FLIP inhibits endothelial cell apoptosis during hyperoxia by suppressing Bax, Free Radical Biol. Med. 42 (2007) 1599–1609.
[39] S.S. Bacus, A.V. Gudkov, M. Lowe, L. Lyass, Y. Yung, A.P. Komarov, K. Keyomarsi, Y. Yarden, R. Seger, Taxol-induced apoptosis depends on MAP kinase pathways (ERK and p38) and is independent of p53, Oncogene 20 (2001) 147–155.
[40] M. Schuler, D.R. Green, Mechanisms of p53-dependent apoptosis, Biochem. Soc. Trans. 7 (2001) 684–688.
[41] C. Gangadharan, M. Thoh, S.K. Manna, Inhibition of constitutive activity of nuclear transcription factor kappaB sensitizes doxorubicin-resistant cells to apoptosis, J. Cell. Biochem. 107 (2009) 203–213.
[42] P. Liu, E. Kimmoun, A. Legrand, A. Sauvanet, C. Degott, B. Lardeux, D. Bernuau, Activation of NF-kappaB, AP-1 and STAT transcription factors is a frequent and early event in human hepatocellular carcinomas, J. Hepatol. 37 (2002) 63–71.
[43] J.E. Chipuk, D.R. Green, P53's believe it or not: Lessons on transcription-independent death, J. Clin. Immunol. 23 (2003) 355–361.
[44] T. Yamanaka, K. Shiraki, K. Sugimoto, Chemotherapeutic agents augment TRAIL-induced apoptosis in human hepatocellular carcinoma cell lines, Hepatology 32 (2000) 482–490.
[45] B. Fischer, D. Coelho, P. Dufour, J.P. Bergerat, J.M. Denis, J. Gueulette, P. Bischoff, Caspase-8-mediated cleavage of the pro-apoptosis Bcl-2 family member Bid in p53-dependent apoptosis, Biochem. Biophys. Res. Commun. 306 (2003) 516–522.
[46] A.R. Safa, T.W. Day, C.H. Wu, Cellular FLICE-like inhibitory protein (C-FLIP): a novel target for cancer therapy, Curr. Cancer Drug Targets 8 (2008) 37–46.
[47] M.T. Dimanche-Boitrel, O. Meurette, A. Rebillard, A. Lacour, Role of early plasma membrane events in chemotherapy-induced cell death, Drug Resist. Update 8 (2005) 5–14.
[48] L. Deng, Y.C. Lin-Lee, F.X. Claret, M.T. Kuo, 2-Acetylaminofluorene upregulates rat mdr 1b expression through generating reactive oxygen species that activate NF-kappa B pathway, J. Biol. Chem. 276 (2001) 413–420.
[49] M.T. Kuo, Z. Liu, Y. Wei, Y.C. Lin-Lee, S. Tatebe, G.B. Mills, H. Unate, Induction of human MDR1 gene expression by 2-acetylaminofluorene is mediated by effectors of the phosphoinositide 3-kinase pathway that activate NF-kappaB signaling, Oncogene 21 (2002) 1945–1954.